

BLE Specs

Birdie PRO

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Purpose of Document

This document provides detailed information about the GATT services and characteristics available on the Birdie Pro BLE device, including UUIDs, value sizes, characteristic properties, and interpretation notes.

1. Generic Access Service (UUID: 1800)

1.1 Device Name

UUID: 2A00

Value Size: Variable

Properties: Read

1.2 Appearance

UUID: 2A01

Value Size: 2 bytes

Properties: Read

2. Birdie Environmental Service (UUID: 4c9b9a7b-370d-473b-8109-f4df6b661012)

2.1 Carbon Dioxide Concentration

UUID: 8fd61dc9-bb74-48c8-924a-d95c57c2f42f

Value Size: 2 bytes

Properties: Read, Notify

Interpretation: uint16_t, ppm

2.2 Temperature

UUID: 2A6E

Value Size: 2 bytes

Properties: Read, Notify

Interpretation: int16_t, millidegrees Celsius (e.g., 22500 = 22.5°C)

2.3 Humidity

UUID: 2A6F

Value Size: 2 bytes

Properties: Read, Notify

Interpretation: uint16_t, milli-%RH (e.g., 45678 = 45.678% RH)

2.4 Birdie state

UUID: 6fdbae35-d241-4a74-bbc3-992010010601

Value Size: 1 byte

Properties: Read, Notify

Interpretation: uint8_t ,**0**-BIRDIE_UP, **1**-BIRDIE_DOWN **2**-BIRDIE_COOLDOWN

3. Birdie Battery Service (UUID: **30b69cc2-03ce-4e58-aa95-b8bffffb4694**)

3.1 Battery Level

UUID: 4769d60c-49b1-4e4b-bc0c-149c5e757bdd

Value Size: 1 byte

Properties: Read, Notify

Interpretation: 0 = Full, 1 = 75%, 2 = 50%, 3 = 25%, 4 = Empty

4. Configuration Service (UUID: **1db703c7-2354-4ffb-bab5-2ae18e414791**)

4.1 Recalibration Threshold

UUID: f7ac6b7e-bb41-4bc3-9fc8-35f73b90f157

Value Size: 2 bytes

Properties: Read, Write

Interpretation: uint16_t, ppm

4.2 Bad IAQ Period

UUID: b2035c1a-07a9-42ce-a2b7-d97c6a34784a

Value Size: 1 byte

Properties: Read, Write

Interpretation: uint8_t, minutes

4.3 IAQ Threshold

UUID: 5f46fade-19a9-4dd5-991e-22729e3ebc44

Value Size: 2 bytes

Properties: Read, Write

Interpretation: uint16_t, ppm

4.4 Cool Down Period

UUID: 2f0c3872-c2be-4416-9610-14a103ed1fc7

Value Size: 1 byte

Properties: Read, Write

Interpretation: uint8_t, minutes

4.5 Good IAQ Period

UUID: 07efb304-17a3-4b77-9adf-777195a3b3ec

Value Size: 1 byte

Properties: Read, Write

Interpretation: uint8_t, minutes

4.6 Firmware Revision String

UUID: 8798982b-8e1a-4102-ab20-b8f1b6a75495

Value Size: 3 bytes

Properties: Read

Interpretation: Byte[0] = major, Byte[1] = minor, Byte[2] = patch

4.7 Averaging window size

UUID: 0fdde7ea-1fe8-48e0-b6c9-9516b3af0233

Value Size: 1 byte

Properties: Read, Write

Interpretation: uint8_t, last N samples used for averaging, MAX value is 12

5. Current Time Service (UUID: 1805)

This is a write-only characteristic used for device time synchronization.

5.1 Current Time

UUID: 00002a2b-0000-1000-8000-00805f9b34fb

Value Size: 4 bytes

Properties: Write Interpretation: uint32_t, Unix Timestamp (seconds since epoch in UTC Little Endian notation)

6. Birdie Data Log Service (UUID: 9daf8c05-ec2b-448d-8c83-ba19f447e9c0)

This service facilitates the transfer and retrieval of historical sensor data.

6.1 Log Data Channel

UUID: 393611dd-4e5f-4023-95ae-623dce1d7242

Value Size: 14 bytes per entry

Properties: Read, Notify

Interpretation: Used for both single index read result and log streaming notifications.

6.2 Log Index Write

UUID: c451887b-6516-48d8-9cb5-4c360bee29b0

Value Size: 2 bytes Properties:

Write Interpretation: uint16_t (**0-8157**) - index of the log entry to read.

6.3 Log Entry Data Structure (14 bytes):

Offset	Size	Type	Field	Description
0	2 bytes	<code>uint16_t</code>	CO2	CO2 concentration in ppm.
2	4 bytes	<code>int32_t</code>	Raw Temp	Temperature in millidegrees Celsius.
6	4 bytes	<code>int32_t</code>	Raw Hum	Humidity in milli-%RH.
10	4 bytes	<code>uint32_t</code>	Timestamp	Unix Timestamp (seconds). 0 if log is empty.

7. Silicon Labs OTA Service (UUID: 1D14D6EE-FD63-4FA1-BFA4-8F47B42119F0)

7.1 Silicon Labs OTA Control

UUID: F7BF3564-FB6D-4E53-88A4-5E37E0326063

Value Size: Variable

Properties: Write